



**CEBU INSTITUTE OF TECHNOLOGY**  
**UNIVERSITY**

# IT342-G01

## SYSTEMS INTEGRATION AND ARCHITECTURE 1

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### **SOFTWARE REQUIREMENTS SPECIFICATION (SRS)**

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Project Title: DormiStay - Dormitory Management System

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### **Table of Contents**

|      |                    |   |
|------|--------------------|---|
| 1.   | Introduction       | 3 |
| 1.1. | Project Title      | 3 |
| 1.2. | Project Background | 3 |
| 1.3. | Problem Statement  | 3 |
| 1.4. | Project Objectives | 3 |

|      |                                                   |                                     |
|------|---------------------------------------------------|-------------------------------------|
| 1.5. | Scope of the Project                              | 3                                   |
| 1.6. | Limitations of the Project                        | 3                                   |
| 2.   | Overall Description                               | 4                                   |
| 2.1. | Product Perspective                               | 4                                   |
| 2.2. | User and Characteristics                          | 4                                   |
| 2.3. | Operating Environment                             | 4                                   |
| 2.4. | Assumptions and Dependencies                      | 5                                   |
| 3.   | System Features and Functional Requirements       | 5                                   |
| 3.1. | Feature 1: Resident Management                    | 5                                   |
| 3.2. | Feature 2: Room Management                        | 5                                   |
| 3.3. | Feature 3: Room Assignment Management             | 5                                   |
| 3.4. | Feature 4: Maintenance Request Management         | 6                                   |
| 3.5. | Feature 5: User Authentication and Access Control | 6                                   |
| 3.6. | Feature 6: Occupancy Monitoring and Reporting     | 6                                   |
| 4.   | Non-Functional Requirements                       | 6                                   |
|      | Business Rules                                    | 7                                   |
|      | Requirement Traceability Table                    | 8                                   |
|      | Revision History Table                            | 8                                   |
| 5.   | System Models (Diagrams)                          | 9                                   |
| 5.1. | Entity +Relationship Diagrams                     | 9                                   |
| 5.2. | Use Case Models                                   | 10                                  |
| 5.3. | Activity Diagram                                  | 11                                  |
| 5.4. | Sequence Diagram                                  | 11                                  |
| 6.   | Appendices                                        | <b>Error! Bookmark not defined.</b> |
| 7.   | Initial Project Timeline                          | 12                                  |
| 8.   | References                                        | 12                                  |
| 9.   | Appendices                                        | 12                                  |

## **1. Introduction**

### **1.1. Project Title**

Dormitory Management Information System

### **1.2. Project Background**

Managing dormitory information manually can be time-consuming and prone to errors. Dormitory administrators need an efficient way to manage resident records, room assignments, maintenance requests, and occupancy information. This system aims to provide a centralized platform for organizing and monitoring dormitory operations.

Guide questions:

- What is the current problem or difficulty?
- Who experiences this problem?
- Why is a software system needed?

### **1.3. Problem Statement**

Dormitory administrators need an efficient way to manage resident records, room assignments, maintenance requests, and occupancy information.

### **1.4. Project Objectives**

- To digitalize dormitory record management.
- To simplify room assignment and resident monitoring.
- To provide an easy way for residents to submit maintenance requests.
- To improve accessibility of dormitory information through web and mobile platforms.

### **1.5. Scope of the Project**

- Resident Management
- Room Management, Assignment & Occupancy Monitoring
- Maintenance Request Management
- Resident Profile Management

### **1.6. Limitations of the Project**

The system will support only one dormitory, basic administrator and resident accounts, room and resident management, and maintenance request tracking. Online payments, real-time notifications, advanced reporting, multi-branch support, and iOS compatibility are outside the scope of the project.

## 2. Overall Description

### 2.1. Product Perspective

The system is a multi-platform system with the property owner and managers maintain the property and units through the web application. The Residents has access to the mobile application for Resident Profile Management and any Maintenance Request.

### 2.2. User and Characteristics

Identify the different types of users and their characteristics.

Dormitory Administrators – Usually the property owner but could be assigned to any hired managers to handle.

Dormitory Residents – The occupants of a unit in the dormitory.

### 2.3. Operating Environment

The DormiStay will operate in a client-server environment consisting of a web application, a mobile application, and a cloud-hosted database.

#### Hardware Requirements

- **Administrator**
  - Desktop or laptop computer
  - Minimum 4 GB RAM
  - Internet connection
- **Resident**
  - Android smartphone
  - Minimum 3 GB RAM
  - Internet connection

#### Software Requirements

- **Users**
  - Modern web browser (Google Chrome, Microsoft Edge, Mozilla Firefox)
  - Android Operating System (Android 8.0 or later)
- **Developer**
  - Java Development Kit
  - Spring Boot Framework
  - ReactJS
  - Android Studio
  - Supabase PostgreSQL Database

#### Development Tools

- IntelliJ IDEA
- VS Code
- Android Studio
- Git and GitHub
- Postman (API Testing)

## 2.4. Assumptions and Dependencies

List any assumptions and external dependencies that may affect the system.

### Assumptions

- Users have access to a stable internet connection.
- Dormitory administrators are responsible for maintaining resident and room information.
- Residents have access to Android mobile devices.
- Users possess basic knowledge of operating web or mobile applications

### Dependencies

- Availability of the Supabase cloud database service.
- Availability of the Spring Boot backend server.
- Compatibility of the system with supported web browsers and Android devices.
- Reliable internet connectivity for communication between the web application, mobile application, backend server, and database.

## 3. System Features and Functional Requirements

Describe each major feature of the system and its functional requirements.

### 3.1. Feature 1: Resident Management

Description: Allows administrators to manage resident information and maintain resident records within the dormitory.

Functional Requirements:

**FR3.1.1** – The system shall allow administrators to add new resident records.

**FR3.1.2** – The system shall allow administrators to view resident information.

**FR3.1.3** – The system shall allow administrators to update resident records.

**FR3.1.4** – The system shall allow administrators to delete resident records.

**FR3.1.5** – The system shall allow residents to view their own profile information.

### 3.2. Feature 2: Room Management

Description: Allows administrators to manage dormitory rooms and their occupancy status.

Functional Requirements:

**FR3.2.1** – The system shall allow administrators to create room records.

**FR3.2.2** – The system shall allow administrators to view room information.

**FR3.2.3** – The system shall allow administrators to update room information.

**FR3.2.4** – The system shall allow administrators to delete room records.

**FR3.2.5** – The system shall display the occupancy status of each room.

### 3.3. Feature 3: Room Assignment Management

Description: Allows administrators to assign residents to available rooms and monitor room occupancy.

Functional Requirements:

- FR3.3.1** – The system shall allow administrators to assign a resident to a room.
- FR3.3.2** – The system shall prevent room assignments that exceed room capacity.
- FR3.3.3** – The system shall allow administrators to transfer residents between rooms.
- FR3.3.4** – The system shall allow administrators to remove room assignments.
- FR3.3.5** – The system shall maintain a record of room assignments.

### **3.4. Feature 4: Maintenance Request Management**

Description: Allows residents to submit maintenance requests and enables administrators to monitor and update request status.

Functional Requirements:

- FR3.4.1** – The system shall allow residents to submit maintenance requests.
- FR3.4.2** – The system shall allow administrators to view maintenance requests.
- FR3.4.3** – The system shall allow administrators to update the status of maintenance requests.
- FR3.4.4** – The system shall allow administrators to mark requests as completed.
- FR3.4.5** – The system shall allow residents to view the status of their requests.

### **3.5. Feature 5: User Authentication and Access Control**

Description: Provides secure access to the system through user authentication and role-based permissions.

Functional Requirements:

- FR3.5.1** – The system shall allow users to log in using registered credentials.
- FR3.5.2** – The system shall allow users to log out securely.
- FR3.5.3** – The system shall restrict access based on user roles.
- FR3.5.4** – The system shall allow administrators to manage user accounts.
- FR3.5.5** – The system shall ensure that residents can only access their own information.

### **3.6. Feature 6: Occupancy Monitoring and Reporting**

Description: Allows administrators to monitor room occupancy and generate basic dormitory reports.

Functional Requirements:

- FR3.6.1** – The system shall display the total number of residents.
- FR3.6.2** – The system shall display occupied and available rooms.
- FR3.6.3** – The system shall provide a summary of maintenance requests.
- FR3.6.4** – The system shall allow administrators to view occupancy reports.
- FR3.6.5** – The system shall allow administrators to search resident and room records.

## **4. Non-Functional Requirements**

Specify system quality attributes such as performance, security, usability, reliability, etc.

### **4.1 Performance**

- NFR4.1.1** – The system shall display requested information within 5 seconds under normal operating conditions.

**NFR4.1.2** – The system shall support multiple users accessing the system simultaneously.

**NFR4.1.3** – The system shall process room assignments and maintenance requests without significant delays.

#### **4.2 Security**

**NFR4.2.1** – The system shall require users to authenticate using valid credentials before accessing protected features.

**NFR4.2.2** – The system shall restrict access to features based on user roles.

**NFR4.2.3** – The system shall protect user information from unauthorized access.

#### **4.3 Usability**

**NFR4.3.1** – The system shall provide a user-friendly interface for both administrators and residents.

**NFR4.3.2** – The system shall maintain consistent navigation across web and mobile platforms.

**NFR4.3.3** – Users shall be able to perform common tasks with minimal training.

#### **4.5 Compatibility Requirements**

**NFR4.5.1** – The web application shall support modern web browsers such as Google Chrome, Microsoft Edge, and Mozilla Firefox.

**NFR4.5.2** – The mobile application shall support Android devices running Android 8.0 or later.

**NFR4.5.3** – The system shall communicate through REST APIs between the web application, mobile application, and backend server.

#### **4.6 Availability Requirements**

**NFR4.6.1** – The system shall be available whenever the backend server and database services are operational.

**NFR4.6.2** – The system shall require an internet connection for data access and synchronization.

### **Business Rules**

**BR1.** Each resident must have a unique resident ID and may only have one active room assignment at a time.

**BR2.** A room shall not exceed its maximum occupancy capacity.

**BR3.** Only available rooms may be assigned to new residents.

**BR4.** Only administrators are authorized to create, update, or delete resident and room records.

**BR5.** Residents may only view and manage their own profile and maintenance requests.

**BR6.** Every maintenance request must be associated with a valid resident and room.

**BR7.** Each maintenance request shall have only one current status (e.g., Pending, In Progress, or Completed).

**BR8.** A resident must be assigned to a room before submitting a maintenance request.

**BR9.** When a resident leaves the dormitory, the room assignment shall be removed, making the room available for future occupants.

**BR10.** The system shall maintain accurate and up-to-date records of residents, rooms, room assignments, and maintenance requests.

### Requirement Traceability Table

| Requirement ID  | Functional Requirement               | Related Use Case            | UML Models                                                |
|-----------------|--------------------------------------|-----------------------------|-----------------------------------------------------------|
| FR3.1.1–FR3.1.5 | Resident Management                  | Manage Residents            | Use Case Diagram, ERD                                     |
| FR3.2.1–FR3.2.5 | Room Management                      | Manage Rooms                | Use Case Diagram, ERD                                     |
| FR3.3.1–FR3.3.5 | Room Assignment Management           | Manage Room Assignments     | Use Case Diagram, ERD                                     |
| FR3.4.1         | Submit Maintenance Request           | Submit Maintenance Request  | Use Case Diagram, ERD, Activity Diagram, Sequence Diagram |
| FR3.4.2–FR3.4.5 | Manage Maintenance Requests          | Manage Maintenance Requests | Use Case Diagram, ERD                                     |
| FR3.5.1–FR3.5.5 | User Authentication & Access Control | Login, Logout               | Use Case Diagram                                          |
| FR3.6.1–FR3.6.5 | Occupancy Monitoring & Reporting     | Generate Occupancy Report   | Use Case Diagram, ERD                                     |

### Revision History Table

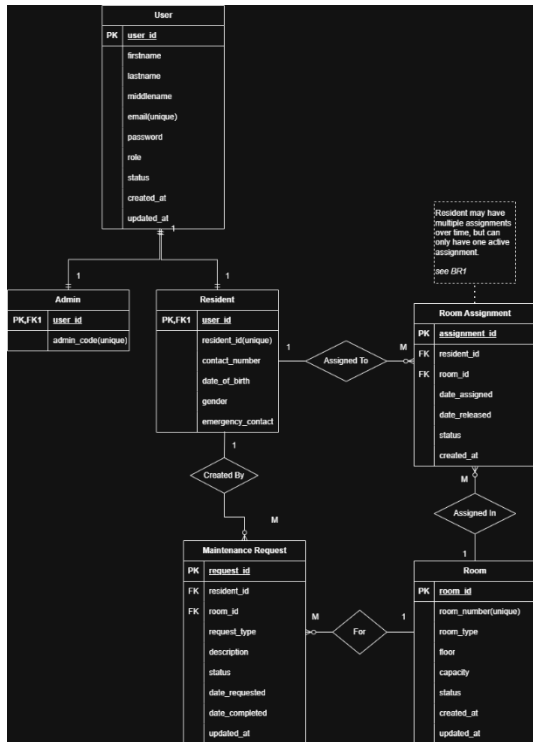
|     |               |                                                          |
|-----|---------------|----------------------------------------------------------|
| 1.0 | June 23, 2026 | Initial SRS Submitted                                    |
| 2.0 | July 07, 2026 | Added UML Models, Created Requirement Traceability Table |



## 5. System Models (Diagrams)

Insert the necessary diagrams for the system:

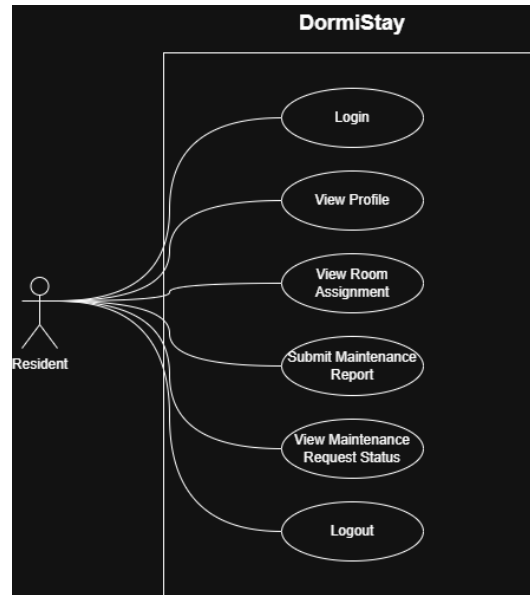
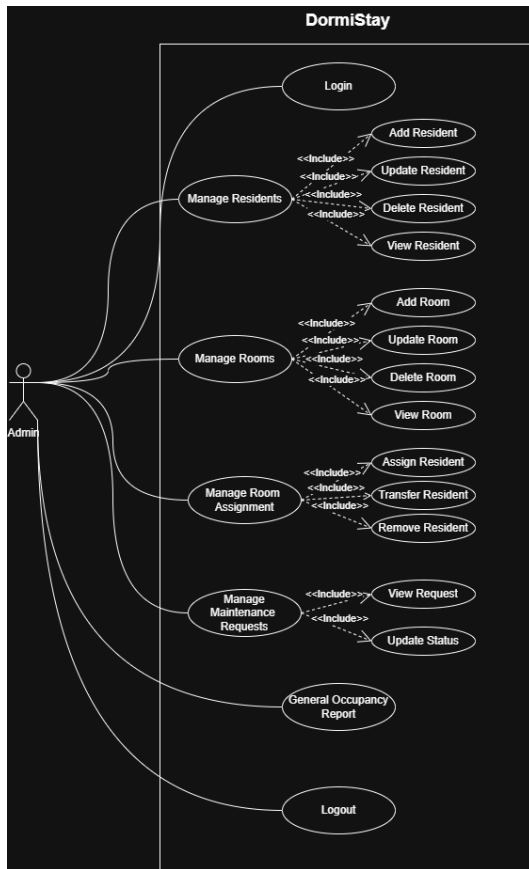
### 5.1. Entity +Relationship Diagrams



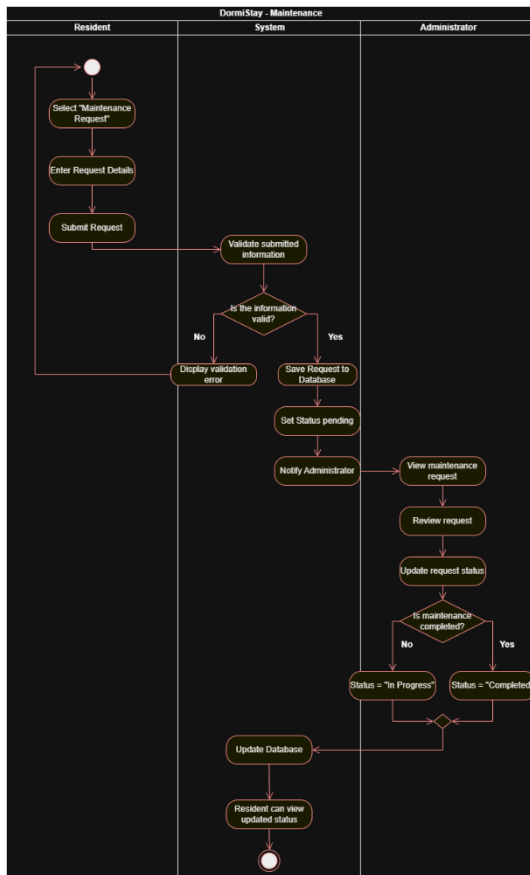
#### Cardinality Summary

- Users 1 – 1 Admin
- User 1 – 1 Resident
- Resident 1 – M Room Assignment
- Room 1 – M Room Assignment
- Resident 1 – M Maintenance Request
- Room 1 – M Maintenance Request

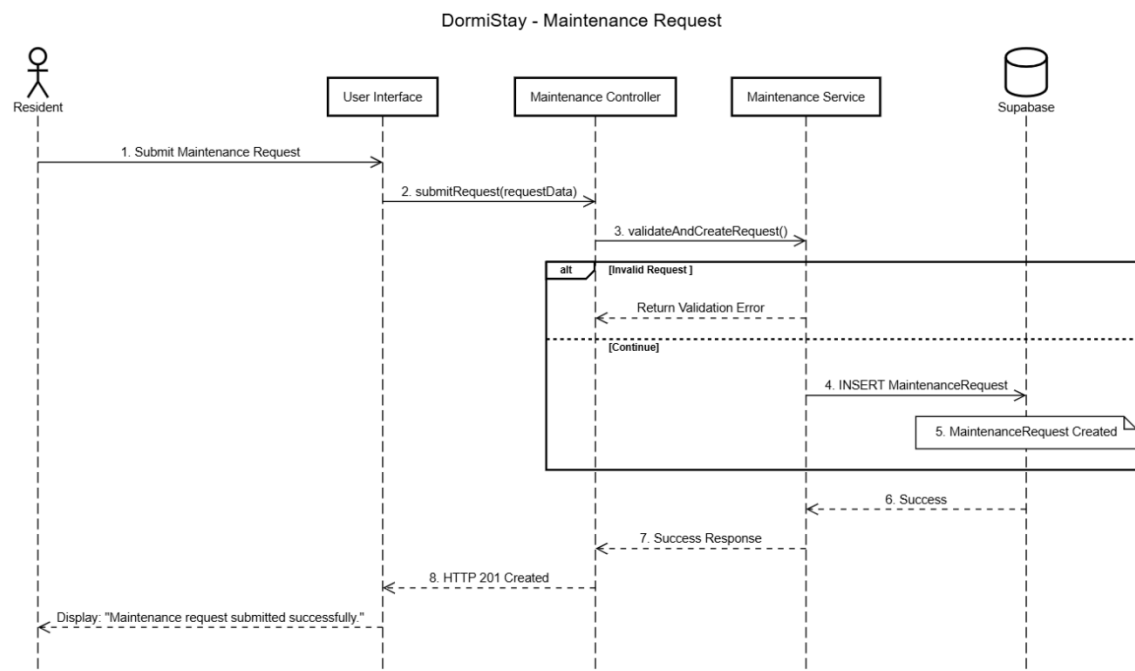
## 5.2. Use Case Models



### 5.3. Activity Diagram



### 5.4. Sequence Diagram



## 6. Initial Project Timeline

| Week   | Planned Activity                |
|--------|---------------------------------|
| Week 1 | Project proposal, Approval, SRS |
| Week 2 | Revise SRS, Register and Login  |
| Week 3 | Develop features                |
| Week 4 | Finalize Integration            |
| Week 5 | Deploy Application              |

## 7. References

- [1] React, "React Learn," *react.dev*, 2024. <https://react.dev/learn> (accessed July 07, 2026)
- [2] Springboot, "Documentation Overview :: Spring Boot," *docs.spring.io*.  
<https://docs.spring.io/spring-boot/documentation.html> (accessed July 07, 2026)
- [3] Supabase, "Supabase Docs," *supabase.com*, 2025. <https://supabase.com/docs> (accessed July 07, 2026)
- [4] "Develop for Android," *Android Developers*. <https://developer.android.com/develop> (accessed July 07, 2026)

## 8. Appendices

No appendices are included in Version of this Software Requirements Specification. Additional supporting documents, such as user interface mockups, wireframes, and supplementary materials, will be provided in future revisions if necessary.